

Donghyun (Daniel) Choi

Scheller College of Business, Georgia Institute of Technology

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EDUCATION

Ph.D. Operations Management, Scheller College of Business **May 2027 (Expected)**
Georgia Institute of Technology

- Minor in Operations Research (School of Industrial and System Engineering)
- Thesis: Essays on Corporate Sustainability Targets (Proposal defended: May 2025)
- Advisors: Dr. Andre Calmon & Dr. Beril Toktay
- Scheller College Ashford Watson Stalnaker Memorial Award for Ph.D. Student Excellence (2026)

M.S. Green Business & Policy, College of Business **Feb. 2017**
Korea Advanced Institute of Science and Technology (KAIST)

- Thesis: Analyzing the Role of the Power Sector Under Korea ETS Using CGE Model.
- Advisor: Dr. Jiyong Eom.
- Highest Honors, Class of 2017 Green Business & Policy Program.

B.S. Urban & Environmental Engineering, **Feb. 2015**
Ulsan National Institute of Science and Technology (UNIST)

- Minor in Finance & Accounting.

EMPLOYMENT

Post-Master Research Associate, **Jun. 2020 - Dec. 2020**
Center for Sustainable Development, KAIST

- Electricity sector CGE (Computable General Equilibrium) modeling and Integrated Assessment Model. (Funded by the Korean Ministry of Environment)

Instructor Officer (1st Lieutenant), **Jun. 2017 - May 2020**
Department of Business Administration, Korea Army Academy

- Selected through a competitive national review of academic record and a subject examination.
- Received the 2018 Best Teaching Award (given to top 5% out of 100 faculty officers)
- Appointment fulfilled the mandatory military service requirement for Korean citizens.

RESEARCH PROJECTS

Working Papers

Choi, D., Shubham, A., Hora, M. “Slack and Science-Based Target Commitments.”

- Under review at *Production and Operations Management* (POM) Journal.
- Presented at *ARCS* (2026, Singapore); *POMS* (2025, Atlanta; 2024, Minneapolis); *MSOM* (2024, Minneapolis).

Choi, D., Calmon, A., Toktay, B. “Closing the Loop or Opening the Market? Implications of Recycled Content Mandate and Allocation Policy.”

- Presented at *MSOM* (2026, Boston); *POMS* (2026, Reno); *INFORMS* (2025, Atlanta; 2024, Seattle); *Novelis Scholars Day* (2025, Novelis Global Research and Technology Center).

Work in Progress

Choi, D., Tang, C., Calmon, A. “Ordered-Quality Process Flexibility in Recycling Networks.”

Shubham, A., Attari, I. **Choi, D.** “Coupled by Commitment? Science-Based Targets and Market-Perceived Climate Risk.”

PUBLICATIONS

Choi, D., Kim, Y.J. (2023) “Local and global experience curves for lumpy and granular energy technologies.” *Energy Policy*, 174, <https://doi.org/10.1016/j.enpol.2023.113426>.

Choi, D., Eom, J., Cho, C.H. (2023) “Integrating Market and Technology Uncertainties into Levelized Costs of Power Generation Technologies: a case of Korea.” *Energy Strategy Reviews*, 45, <https://doi.org/10.1016/j.esr.2022.101046>.

Choi, D., Ahn, Y.H., Choi, D.G. (2020) “Multi-Criteria Decision Analysis of Electricity Sector Transition Policy in Korea.” *Energy Strategy Reviews*, 29, <https://doi.org/10.1016/j.esr.2020.100485>.

Kim, J.T., Cho, C.H., **Choi, D.** (2019). “A CGE Analysis of Peak Load Pricing using Electricity Sector Detailed Database.” *Korean Energy Economic Review*, 18(2): 1–37. [Link](#) (in Korean)

Choi, D., Kasdan, D.O., Yoon, D.K. (2019) “Analyzing Disaster Loss Trends: A Comparison of Normalization Methodologies in South Korea.” *Risk Analysis*, 39(4): 859–870. [Link](#)

OTHER PUBLICATIONS

Toktay, B., Shubham, A., **Choi, D.**, Hora, M. (2024, November 20). “Companies are still committing to net-zero emissions, even if it’s a bumpy road – here’s what the data show” *The Conversation*

TEACHING

Instructor of Record, Georgia Institute of Technology

MGT3501, Operations Management, UG

• Spring 2027 (In-person)	Scheduled
• Spring 2026 (In-person, 60 students)	4.77/5.00
• Summer 2025 (Online, 59 students)	4.63/5.00
• Spring 2025 (In-person, 60 students)	4.65/5.00

Instructor of Record, Korea Army Academy (2017 - 2020)

• Business Statistics, UG	6 Sections
• Management Science, UG	2 Sections
• Decision Analysis, UG	2 Sections
• Financial Accounting, UG	2 Sections
• Managerial Accounting, UG	2 Sections

Received the 2018 Best Teaching Award (given to top 5% out of 100 faculty officers)

GRANTS, FELLOWSHIPS, AND AWARDS

- **Ashford Watson Stalnaker Memorial Award for Ph.D. Student Excellence** (2026) \$5,000
College-level award recognizing exemplary scholarship and innovation in both research and teaching.
- **Novelis Graduate Scholar** (2025) \$8,000
Competitive award to develop a research project aligned with Novelis’ Vision 3x30 objectives on recycled content, CO₂e intensity reduction, and return on invested capital. [Link](#)

- **Sheth Fellows Award** (2024) \$1,000
Awarded to Ph.D. students who have reached a major research milestone. [Link](#)
- **BBISS (Brook Byers Institute for Sustainable Systems) Microgrant** (2023) \$500
Seed grant awarded on the basis of a competitive research proposal. [Link](#)
- **Winner, 2023 CSCN (Circular Supply Chain Network) Hackathon** (2023) \$3,200
First place out of 19 participating teams.

DOCTORAL AND GRADUATE COURSEWORK

- **Optimization and Stochastic Modeling:** Deterministic Optimization, Nonlinear Optimization, Stochastic Processes, Probabilistic Models, Simulation, Mathematical Programming[†]
- **Econometrics and Statistical Methods:** Econometrics, Multivariate Data Analysis, Machine Learning
- **Operations & Economics:** Operations Management[†], Microeconomic Theory II, Energy & Environmental Economics[†], Industrial Organization[†]

[†]Completed in the M.S. program at KAIST; all others in the Ph.D. program at Georgia Tech.

SERVICE

- Invited Session Chair, POMS Annual Conference (2025, 2026)
- Student Volunteer, POMS Annual Conference (2024, 2025)
- Reviewer, Production and Operations Management (POM) Journal

REFERENCES

Andre Calmon (Co-Chair)

Associate Professor of Operations Management
Faculty Director, Ray C. Anderson Center for Sustainable Business
Scheller College of Business, Georgia Tech
andre.calmon@gatech.edu

Manpreet S. Hora

Senior Associate Dean for Programs
Dean's Distinguished Professor
Scheller College of Business, Georgia Tech
manpreet.hora@scheller.gatech.edu

Jiyong Eom (M.S. Advisor)

Professor, Business & Technology Management,
KAIST College of Business
eomjiyong@kaist.ac.kr

L. Beril Toktay (Co-Chair)

Brady Family Chair and Regents' Professor
Executive Director, Brook Byers Institute for Sustainable Systems
Scheller College of Business, Georgia Tech
beril.toktay@scheller.gatech.edu

Benjamin R. Jordan

Fmr. Global Lead, Packaging & Circular Economy, The Coca-Cola Company
Adjunct Lecturer, Carter School of Public Policy, Georgia Tech
bjordan3@gatech.edu